

Prime-Frequency Neural Inverse Kinematics versus Halton–Prime Configuration-Space Planning from 5 to 30 DOF

L.A. Muñoz

Computational Robotics Lab
Tec de Monterrey

July 2026

Abstract

We study collision-free motion generation for planar redundant manipulators whose number of degrees of freedom (DOF) is swept from $N = 5$ to $N = 30$ while the total reach, the workspace, the obstacles, and the task are held *exactly* constant, so that dimensionality is the only experimental variable. Two families of methods are compared head to head: **Method A**, a GPU-parallel population of neural inverse-kinematics (IK) policies whose inputs are quantized through sinusoidal channels at *distinct prime frequencies* per joint, and **Method B**, a classical configuration-space pipeline (damped least-squares IK followed by RRT-Connect) whose random sampling is replaced by the deterministic, low-discrepancy *Halton sequence with one distinct prime base per joint*. The unifying idea is that the prime numbers provide a canonical, inexhaustible supply of pairwise-coprime integers: adding a degree of freedom simply means consuming the next prime, which yields (i) input encodings whose channels share no common harmonics and (ii) space-filling samplers whose coordinates remain decorrelated as the dimension grows. We formalize both mechanisms, give the complete kinematic, collision, loss, and planning equations implemented in the accompanying experiment suite, and describe the robustness machinery (near-zero policy initialization, absolute self-collision barriers, population-wide feasibility selection, and feasible-snapshot polishing) required to make the neural method scale to 30 DOF. We discuss why prime-based constructions are an elegant alternative for growing dimensionality: the free fraction of the configuration space collapses roughly exponentially with N , and coverage quality — exactly what coprimality buys — is what matters most in that regime.

1 Introduction

Redundant manipulators trade kinematic simplicity for dexterity: every additional joint enlarges the null space available for obstacle avoidance, but it also inflates the configuration space (C-space) whose free subset must be searched or learned. Two broad strategies dominate. *Learning-based inverse kinematics* amortizes the cost of motion generation into a trained network, but requires an input representation that remains informative as the joint vector grows. *Sampling-based planning* [3, 4] attacks the C-space directly, but its performance is governed by how well a finite set of samples covers a high-dimensional cube whose free fraction shrinks rapidly.

This paper develops and evaluates a single design principle that serves both strategies at once: **index every degree of freedom by a distinct prime number**. Concretely,

- (i) in the neural method, joint k (and each Cartesian coordinate) is expanded into $\sin / \cos$ channels oscillating at the k -th prime frequency, so that no two channels share a harmonic (Section 3);
- (ii) in the planner, the k -th coordinate of every sample is generated by the Halton radical-inverse in base p_k , the k -th prime, producing a deterministic low-discrepancy point set whose dimensions are decorrelated by the pairwise coprimality of the bases (Section 3.2).

The appeal is structural as much as numerical. The primes form an infinite, canonical, pairwise-coprime sequence; therefore *scaling the robot from N to $N + 1$ degrees of freedom requires no re-design of the representation whatsoever* — one simply appends the next prime. No hand-tuned frequency schedule, no re-drawn random projection, no re-balanced lattice: the arithmetic of $\mathbb{Z}$ supplies the extension for free. We argue in Section 3.3 that this makes primes an unusually elegant answer to the question “how should a representation grow with the number of joints?”

To isolate the effect of dimensionality, the experimental suite fixes the total reach of the arm at $L = 10$ (so the link length is L/N), and keeps the start pose, goal, and obstacle field identical across $N \in \{5, 10, 15, 20, 25, 30\}$. Everything that changes between experiments is a consequence of N alone.

Contributions. (1) A fully differentiable, batched planar N -DOF arm model with per-segment external clearance and segment–segment self-collision clearance (Section 2). (2) Method A: a GPU population of prime-encoded IK policies with continuity guaranteed by construction, together with the four robustness mechanisms empirically required at 20–30 DOF (Section 4). (3) Method B: a DLS + RRT-Connect pipeline whose sampler is the prime-base Halton sequence, benchmarked against uniform pseudo-random sampling at every DOF (Section 5). (4) A common measurement protocol — free-space fraction, accuracy, tracking, continuity, clearances, wall time, and a hardware-agnostic “token” cost — applied uniformly across the sweep (Section 6).

2 Problem Formulation and Arm Model

2.1 Kinematics with constant total reach

Consider a planar serial arm with N revolute joints, joint vector $\mathbf{q} = (q_1, \dots, q_N) \in \mathcal{Q} = [-q_{\text{lim}}, q_{\text{lim}}]^N$ with $q_{\text{lim}} = 0.9\pi$ ($\pm 162^\circ$), and *constant total reach*

$$L = 10, \quad \ell = \frac{L}{N}, \quad r_{\text{link}} = 0.08 \ell. \quad (1)$$

The cumulative link orientations and joint positions are

$$\phi_k = \sum_{j=1}^k q_j, \quad \mathbf{p}_0 = \mathbf{0}, \quad \mathbf{p}_k = \mathbf{p}_{k-1} + \ell \begin{pmatrix} \cos \phi_k \\ \sin \phi_k \end{pmatrix}, \quad k = 1, \dots, N, \quad (2)$$

and the end effector is $\mathbf{e}(\mathbf{q}) = \mathbf{p}_N$. Equation (2) is implemented as two vectorized cumulative sums, so forward kinematics is differentiable and evaluable for arbitrary batch shapes on the GPU. The analytical end-effector Jacobian $J(\mathbf{q}) \in \mathbb{R}^{2 \times N}$ has columns

$$J_{:,i}(\mathbf{q}) = \begin{pmatrix} -(e_y - p_{i-1,y}) \\ (e_x - p_{i-1,x}) \end{pmatrix}, \quad i = 1, \dots, N, \quad (3)$$

i.e. the standard planar revolute column: a 90° rotation of the lever arm from joint i to the tip.

The shared scenario places the start pose “straight up” ($q_1 = 90^\circ$, $q_{k>1} = 0$), so the end effector starts at $\mathbf{e}_0 = (0, L) = (0, 10)$ for *every* N ; the goal is $\mathbf{g} = (6, -6)$; and three circular obstacles $\{(\mathbf{c}_m, r_m)\}_{m=1}^3$ with centers $(5, 1)$, $(6, -2.5)$, $(2.5, -4)$ and radii 1.3, 1.1, 0.9 populate the workspace. A safety margin $\delta = 0.25$ is enforced against external obstacles.

2.2 Collision model

External clearance. Each link is the segment $[\mathbf{p}_{k-1}, \mathbf{p}_k]$. Its clearance to obstacle m is the point–segment distance minus the obstacle radius:

$$d_{k,m}^{\text{ext}}(\mathbf{q}) = \left\| \mathbf{p}_{k-1} + t_{k,m}^* (\mathbf{p}_k - \mathbf{p}_{k-1}) - \mathbf{c}_m \right\| - r_m, \quad t_{k,m}^* = \text{clip} \left(\frac{(\mathbf{c}_m - \mathbf{p}_{k-1})^\top (\mathbf{p}_k - \mathbf{p}_{k-1})}{\|\mathbf{p}_k - \mathbf{p}_{k-1}\|^2}, 0, 1 \right). \quad (4)$$

Self-collision clearance. For every non-adjacent link pair (i, j) with $j \geq i+2$, let $d_{\text{seg}}([\mathbf{p}_{i-1}, \mathbf{p}_i], [\mathbf{p}_{j-1}, \mathbf{p}_j])$ denote the exact minimum segment–segment distance (computed by the standard clamped closest-point parametrization, fully vectorized over batches and pairs). The internal clearance is

$$d_{i,j}^{\text{self}}(\mathbf{q}) = d_{\text{seg}}([\mathbf{p}_{i-1}, \mathbf{p}_i], [\mathbf{p}_{j-1}, \mathbf{p}_j]) - 2r_{\text{link}}. \quad (5)$$

The number of pairs grows quadratically, $\binom{N-1}{2} - (N-2)$ non-adjacent pairs, which is one of the two mechanisms by which the problem hardens with N .

Free-configuration predicate. A configuration is free iff all clearances and limits hold simultaneously:

$$F(\mathbf{q}) = \left[\min_{k,m} d_{k,m}^{\text{ext}}(\mathbf{q}) > 0 \right] \wedge \left[\min_{i,j} d_{i,j}^{\text{self}}(\mathbf{q}) > 0 \right] \wedge \left[\|\mathbf{q}\|_\infty \leq q_{\text{lim}} \right]. \quad (6)$$

Trajectories $\mathbf{q}_{0:T}$ are certified by *dense validation*: each consecutive pair is linearly interpolated with 5 sub-steps and (6) is evaluated on the refined sequence, returning the boolean verdict together with the minimum external and internal clearances encountered.

2.3 Shared Cartesian reference path

Because the end-effector endpoints are DOF-independent, a single obstacle-free Cartesian path $\{\mathbf{w}_t\}_{t=0}^{T-1}$, $T = 60$, is optimized once and reused by both methods and all DOF. With the endpoints pinned ($\mathbf{w}_0 = \mathbf{e}_0$, $\mathbf{w}_{T-1} = \mathbf{g}$), the interior waypoints solve

$$\min_{\mathbf{w}_{1:T-2}} \sum_{t=0}^{T-2} \|\mathbf{w}_{t+1} - \mathbf{w}_t\|^2 + 80 \sum_t \sum_m \text{ReLU} \left((r_m + \delta + 0.15) - \|\mathbf{w}_t - \mathbf{c}_m\| \right)^2, \quad (7)$$

minimized by Adam for 600 iterations. The result is a short, smooth path that clears every obstacle by at least $\delta + 0.15$.

3 The Prime-Number Machinery

For an N -DOF arm the suite consumes the first $N + 2$ primes $p_1 < p_2 < \dots < p_{N+2}$: the first N are assigned one-per-joint, and the last two to the Cartesian coordinates of the target.

3.1 Prime-frequency input quantization (Method A)

Given a vector $\mathbf{v} \in \mathbb{R}^d$, its assigned primes $(p_{k_1}, \dots, p_{k_d})$ and a scale s , the *prime encoding* is

$$\psi(\mathbf{v}; s) = \left[\sin\left(\frac{p_{k_1} v_1}{s}\right), \dots, \sin\left(\frac{p_{k_d} v_d}{s}\right), \cos\left(\frac{p_{k_1} v_1}{s}\right), \dots, \cos\left(\frac{p_{k_d} v_d}{s}\right) \right] \in \mathbb{R}^{2d}. \quad (8)$$

The full feature vector consumed by the policy at rollout step t , with target waypoint $\mathbf{w}_t$, previous configuration $\mathbf{q}_{t-1}$, and normalized progress $\tau = t/(T-1)$, is

$$\mathbf{x}_t = \left[\mathbf{w}_t; \psi(\mathbf{w}_t; L); \mathbf{q}_{t-1}; \psi(\mathbf{q}_{t-1}; q_{\text{lim}}); \tau, \sin \pi \tau, \cos \pi \tau \right] \in \mathbb{R}^{3N+9}. \quad (9)$$

The feature dimension grows only *linearly* in N : appending a joint appends one raw angle and one (sin, cos) pair at the next prime frequency.

3.2 Prime-base Halton sampling (Method B and C-space estimation)

Write the index $i \in \mathbb{N}$ in base b as $i = \sum_{j \geq 0} a_j(i) b^j$. The *radical inverse* mirrors the digits about the radix point,

$$h_b(i) = \sum_{j \geq 0} a_j(i) b^{-(j+1)} \in [0, 1), \quad (10)$$

and the N -dimensional Halton sequence [1] with the prime bases assigned to the joints is

$$\mathbf{u}_i = (h_{p_1}(i), h_{p_2}(i), \dots, h_{p_N}(i)) \in [0, 1)^N, \quad \mathbf{q}_i = (2\mathbf{u}_i - \mathbf{1}) q_{\text{lim}}. \quad (11)$$

Halton points are deterministic and *low-discrepancy*: their star discrepancy satisfies

$$D_M^*(\mathbf{u}_1, \dots, \mathbf{u}_M) = O\left(\frac{(\log M)^N}{M}\right), \quad (12)$$

compared with the $O(M^{-1/2})$ statistical fluctuation of i.i.d. uniform sampling [2]. The pairwise coprimality of the bases is not decorative — it is the property that prevents pairs of coordinates from locking onto a common cycle: if two bases shared a factor, the corresponding two-dimensional projection of the sequence would concentrate on a coarse sublattice.

3.3 Why primes? An elegant scaling principle

Both mechanisms above are instances of one requirement: *a family of integers, one per DOF, that is pairwise coprime and extends indefinitely*. The primes are the canonical — indeed the minimal and universal — such family, and this is the conceptual center of the paper. Four properties make the choice more than a convenience.

(a) No shared harmonics. In the encoding (equation (8)), channel k oscillates at integer frequency p_k (in units of $1/s$). Two channels alias — produce indistinguishable responses on a common sub-period — exactly when their frequency ratio is a ratio of small integers. For distinct primes, p_i/p_j is never an integer and the pair (p_i, p_j) has least common multiple $p_i p_j$, the largest possible; hence the joint response $(\sin p_i \theta, \sin p_j \theta)$ traverses its longest possible pattern before repeating. Contrast this with the geometric (powers-of-two) frequency schedules popular in positional encodings [7, 8]: there every frequency divides all higher ones, so entire towers of channels share harmonics by construction. The prime schedule is the antithesis — *maximally incommensurate integer frequencies* — and it hands each joint a spectrally private channel.

(b) Scaling by appending. When the arm grows from N to $N + 1$ joints, the representation is extended by the single datum p_{N+1} . Nothing already learned or already assigned must be revisited: the frequencies of joints $1, \dots, N$ are unchanged, and the new channel is automatically coprime to all of them. The same statement holds verbatim for the Halton bases in (equation (11)). In this sense the prime sequence converts “increase the number of degrees of freedom” — normally an invasive architectural operation — into the cheapest possible incremental one: *take the next prime*. This is the sense in which we call the construction elegant: the difficulty of scaling is absorbed by a pre-existing, infinitely extensible object of pure arithmetic rather than by engineering.

(c) Determinism and reproducibility. Unlike random Fourier features or random planners, both prime constructions are fully deterministic. Every run of the suite, on any hardware, draws the *same* Halton samples and encodes with the *same* frequencies; sampler comparisons are therefore exactly repeatable, and the uniform pseudo-random baseline isolates precisely the contribution of low-discrepancy coverage.

(d) Coverage where it matters. Section 6 recalls that the free fraction of the C-space collapses roughly exponentially with N under constant reach. In that regime the value of a sampler is measured by how uniformly a *small* number of points covers a cube whose interesting subset is thin: exactly the property quantified by (12). The suite makes this concrete by estimating the free fraction with both samplers and by running RRT-Connect with both samplers at every DOF.

Honest caveats. Two limitations temper the elegance. First, large primes mean large frequencies: at high N the last channels of (8) oscillate rapidly over the input range, which can make them noisy features unless the scale s is chosen per-magnitude, as done here ($s = L$ for Cartesian inputs, $s = q_{\text{lim}}$ for angles). Second, the classical Halton sequence is known to develop correlated stripe patterns in projections onto pairs of *high-index* (large-base) dimensions for small sample counts; scrambling variants mitigate this. The experiment suite deliberately uses the plain construction to keep the prime story pure, and reports the uniform baseline at every DOF so that any degradation is visible rather than assumed.

4 Method A: Population of Prime-Encoded Neural IK Policies

4.1 Policy, rollout, and continuity by construction

A population of B networks (64 on GPU) is trained fully in parallel: each layer is a *batched* linear map with one weight slab per member, evaluated by a single `bmm`. Each member is a three-layer GELU MLP

$$f_{\theta}(\mathbf{x}) = \tanh\left(W_3 \sigma(W_2 \sigma(W_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3\right) \cdot \Delta_{\max}, \quad \Delta_{\max} = 8^\circ, \quad (13)$$

with hidden width 256 (GPU). The bounded output makes the trajectory *continuous by construction*:

$$\mathbf{q}_t = \text{clip}(\mathbf{q}_{t-1} + f_{\theta}(\mathbf{x}_t), -q_{\text{lim}}, q_{\text{lim}}), \quad \|\mathbf{q}_t - \mathbf{q}_{t-1}\|_{\infty} \leq \Delta_{\max}, \quad (14)$$

for $t = 1, \dots, T - 1$, starting at the shared start pose $\mathbf{q}_0$.

Near-zero output initialization. The output layer is initialized at 2% of its Xavier scale, so the initial policy is “hold still.” This was found to be *decisive* at 20–30 DOF: without it, early rollouts tangle the arm into self-crossing homotopy classes from which gradient descent cannot escape (see the barrier discussion below).

4.2 Training loss and curriculum

Let $\mathbf{q}_{0:T-1}^{(b)}$ be member b 's rollout. Its per-member loss is

$$\begin{aligned} \mathcal{L}^{(b)} = & \underbrace{\sum_{t=1}^{T-1} \|\mathbf{e}(\mathbf{q}_t^{(b)}) - \mathbf{w}_t\|^2}_{\text{Cartesian tracking}} + 60 \underbrace{\|\mathbf{e}(\mathbf{q}_{T-1}^{(b)}) - \mathbf{w}_{T-1}\|^2}_{\text{goal accuracy}} + w_{\text{ext}}(e) P_{\text{ext}}^{(b)} + 60 P_{\text{self}}^{(b)} \\ & + 0.10 \sum_t \|\Delta \mathbf{q}_t^{(b)}\|^2 + 0.30 \sum_t \|\Delta^2 \mathbf{q}_t^{(b)}\|^2, \end{aligned} \quad (15)$$

where Δ and Δ^2 are first and second temporal differences (velocity and jerk surrogates), and the penalties hinge on the clearances (4)–(5):

$$P_{\text{ext}} = \sum_{t,k,m} \text{ReLU}(\delta - d_{k,m}^{\text{ext}}(\mathbf{q}_t))^2, \quad (16)$$

$$P_{\text{self}} = \sum_{t,i,j} \text{ReLU}(m_s - d_{i,j}^{\text{self}}(\mathbf{q}_t))^2, \quad m_s = \max(0.15, 2r_{\text{link}}). \quad (17)$$

Only the *external* weight follows a curriculum,

$$w_{\text{ext}}(e) = 2 + 28 \min(1, e/(0.6E)), \quad (18)$$

ramping from 2 to 30 over the first 60% of the E epochs, while the self-collision barrier is at full strength from epoch 0. The asymmetry is deliberate: two links at distance zero (crossed) yield a *symmetric* penalty landscape with no gradient toward uncrossing, so crossings must be *prevented*, never repaired — hence the constant barrier, the absolute margin m_s in (17) (a purely radius-scaled margin lets short links cross at high N), and the hold-still initialization.

Optimization uses Adam (learning rate 3×10^{-3}), cosine-annealed, with gradient-norm clipping at 5. The epoch budget scales with the genuinely harder landscapes:

$$E_N = E \cdot \left(1 + \frac{\max(0, N-10)}{20}\right), \quad E = 1500 \text{ (GPU)}. \quad (19)$$

Population-wide feasibility selection. Every 5 epochs (after a 20% warm-up) the *exact* predicate (6) is evaluated over the entire population in one batched GPU call; among the members whose whole rollout is free, the lowest-loss one is densely validated, and only then may it become the incumbent trajectory. Feasibility, not loss, is the gate.

4.3 Feasible-snapshot polishing

The incumbent (and up to five lowest-loss candidates) is refined by directly optimizing the bounded increments. Reparametrize $\mathbf{r}_t = \text{artanh}((\mathbf{q}_{t+1} - \mathbf{q}_t)/\Delta_{\text{max}})$ so that

$$\mathbf{q}_t = \mathbf{q}_0 + \sum_{t' < t} \tanh(\mathbf{r}_{t'}) \Delta_{\text{max}} \quad (20)$$

keeps the 8° bound intact for free, and minimize

$$\mathcal{L}_{\text{pol}} = \sum_t \|\mathbf{e}(\mathbf{q}_t) - \mathbf{w}_t\|^2 + 300 \|\mathbf{e}(\mathbf{q}_{T-1}) - \mathbf{g}\|^2 + w_{\text{pen}}(i) \left(P_{\text{ext}} + 3 P_{\text{self}}^{(m'_s)} \right) + 0.1 \sum_t \|\Delta \mathbf{q}_t\|^2 + 0.3 \sum_t \|\Delta^2 \mathbf{q}_t\|^2, \quad (21)$$

with a reduced internal margin $m'_s = \max(0.10, r_{\text{link}})$ and a penalty ramp $w_{\text{pen}}(i) = 50 + 350 \min(1, i/(0.5 I))$ over $I = 800$ iterations. Every 25 iterations the current trajectory is densely validated and the feasible snapshot with the lowest final error is retained; polishing therefore *can only improve a feasible input, never destroy its feasibility*. The Method-A “token” count accumulates every scalar feature consumed during training plus every scalar optimized during polishing.

5 Method B: Configuration-Space Planning with Halton–Prime Sampling

5.1 Damped least-squares tracking with avoidance

A continuous reference to the goal configuration is produced by damped least-squares (DLS) IK [5, 6] along the shared Cartesian path. With task error $\mathbf{e}_{\text{task}} = \mathbf{w} - \mathbf{e}(\mathbf{q})$ clamped to $\|\mathbf{e}_{\text{task}}\| \leq 0.3$, the raw update is

$$\Delta \mathbf{q}_{\text{DLS}} = \mathbf{J}^\top (\mathbf{J} \mathbf{J}^\top + \lambda^2 \mathbf{I}_2)^{-1} \mathbf{e}_{\text{task}}, \quad \lambda = 0.15, \quad (22)$$

augmented by the negative gradient of a differentiable avoidance potential,

$$\Phi(\mathbf{q}) = \sum_{k,m} \text{ReLU}\left(a_{\text{ext}} - (d_{k,m}^{\text{ext}}(\mathbf{q}) - \delta)\right)^2 + \sum_{i,j} \text{ReLU}\left(1.5 r_{\text{link}} - d_{i,j}^{\text{self}}(\mathbf{q})\right)^2, \quad a_{\text{ext}} = \min(0.30, 0.30 \ell), \quad (23)$$

(the activation band scales with the link length; a fixed band stalls short-link arms), yielding the micro-step

$$\Delta \mathbf{q} = \Delta \mathbf{q}_{\text{DLS}} - 0.25 \nabla_{\mathbf{q}} \Phi(\mathbf{q}), \quad \|\Delta \mathbf{q}\|_\infty \leq 2^\circ. \quad (24)$$

The 2° micro-step bound makes the DLS trajectory continuous by construction. Its terminus $\mathbf{q}_{\text{goal}}$ is, if necessary, repaired by a short Adam optimization of

$$200 \|\mathbf{e}(\mathbf{q}) - \mathbf{g}\|^2 + 30 \sum \text{ReLU}(0.05 - d^{\text{ext}})^2 + 30 \sum \text{ReLU}(0.02 - d^{\text{self}})^2 \quad (25)$$

until it is exactly on the goal and strictly free.

5.2 RRT-Connect with prime-base Halton samples

RRT-Connect [4] then plans in $\mathcal{Q}$ between $\mathbf{q}_0$ and $\mathbf{q}_{\text{goal}}$ with dimension-aware parameters:

$$\eta = 15^\circ \sqrt{N} \quad (\text{extension step}), \quad I_{\text{max}} = 3000 + 200 N \quad (\text{iteration budget}), \quad (26)$$

every candidate edge checked at 2° resolution with an extra clearance margin of 0.05 (with constant total reach, a joint step of $d\mathbf{q}$ can move the tip by up to $L d\mathbf{q}$ regardless of N , so edge resolution must not be relaxed with dimension). The random configuration at iteration i is drawn either from the Halton table (11) (8000 precomputed points, cycled) — the *prime* sampler — or from a seeded uniform generator — the baseline. Both samplers are run at every DOF with identical budgets, and the planner records success, iterations, tree size, wall time, and the exact number of collision-checker queries.

Successful paths are post-processed by 400 iterations of random shortcutting (an edge (i, j) replaces the intermediate nodes whenever it is collision-free) and then resampled so that every step obeys $\|\Delta \mathbf{q}\|_\infty \leq 3^\circ$ — continuity by construction on the planner side as well — followed by dense validation. If RRT exhausts its budget (possible at very high N , where the free fraction

has collapsed), the already-continuous, already-validated DLS trajectory is used as the Method-B output and the failure is recorded rather than hidden. The Method-B token count is

$$\text{tokens}_B = (\#\text{queries}_{\text{RRT}} + \#\text{queries}_{\text{DLS}}) N + |\text{DLS trajectory}| \cdot N, \quad (27)$$

i.e. every configuration touched by the collision checker, counted in scalars, on the same footing as Method A’s feature scalars.

6 Experimental Protocol and Metrics

6.1 C-space geometry as the common backdrop

Before any planning, the free fraction of $\mathcal{Q}$ is estimated at every DOF with $M = 60,000$ GPU-batched evaluations of (6) per sampler,

$$\hat{\rho} = \frac{1}{M} \sum_{i=1}^M \mathbf{1}[F(\mathbf{q}_i)], \quad \mathbf{q}_i \text{ from (11) (Halton-prime) or i.i.d. uniform,} \quad (28)$$

while the raw C-space volume grows as $(2q_{\text{lim}})^N$. Because every additional non-adjacent link pair in (5) is a fresh opportunity to self-collide, $\hat{\rho}$ collapses roughly exponentially in N under constant total reach — this measured collapse is the geometric backdrop against which both methods are read, and the low-variance behavior of the Halton estimate at fixed budget is itself a first, planner-independent illustration of the value of prime-base coverage.

6.2 Per-method metrics

For each (N, method) pair the suite records:

- **Final error** $\|\mathbf{e}(\mathbf{q}_{\text{final}}) - \mathbf{g}\|$;
- **Tracking error**: the mean distance $\frac{1}{T'} \sum_t \|\mathbf{e}(\mathbf{q}_t) - \tilde{\mathbf{w}}_t\|$ to the reference path linearly resampled to the trajectory length T' ;
- **Continuity**: $\max_t \|\mathbf{q}_{t+1} - \mathbf{q}_t\|_\infty$, verified against the hard bounds (8° for A, 3° for B) at every DOF;
- **Clearances**: minimum external and internal clearance over the densely interpolated trajectory, and the boolean dense-validation verdict;
- **Cost**: wall time and the token counts defined above — a hardware-agnostic proxy for computation on either side;
- **Sampler comparison** (Method B only): success, iterations, and collision queries of Halton-prime versus uniform at identical budgets.

All figures (workspace, C-space collapse, training convergence, joint profiles, per-step continuity, and a six-panel final comparison), one side-by-side animation per DOF, and a machine-readable CSV are emitted by the suite.

7 Results and Discussion

The accompanying notebook is the executable form of this paper; the quantitative values (errors, times, token counts, sampler iterations) are produced per run and per hardware. The qualitative findings the suite is designed to exhibit — and which its conclusions record — are the following.

The geometry hardens exponentially, the task does not. Holding reach, workspace, obstacles, and endpoints fixed while sweeping N cleanly separates *task difficulty* (constant) from *search-space difficulty* (exploding). The free fraction (28) decays roughly exponentially with N , dominated by the quadratically growing family of self-collision pairs. Any method that scales must therefore either learn structure (A) or cover space efficiently (B) — and the prime constructions serve precisely these two needs.

Method A scales gracefully with the prime encoding, given the right barriers. Per-epoch cost grows mildly (features are $O(N)$ by (9); self-collision terms are $O(N^2)$ but fully batched), and the GPU keeps the whole population parallel. Prime quantization keeps the encoding informative as the input dimension grows: every joint owns an incommensurate frequency, so the representation does not degenerate into mutually aliased channels the way a shared or harmonic frequency schedule would. Continuity is bounded at 8° by construction (14) at every DOF, and polishing drives the final error to the order of 10^{-2} across the sweep. Crucially, the four robustness mechanisms of Section 4 — hold-still initialization, constant absolute self-collision barrier, batched feasibility selection, and feasible-snapshot polishing — were found empirically to be what separates “works at 10 DOF” from “works at 30 DOF.”

Method B is cheap at low DOF; prime-base Halton earns its keep as N grows. DLS + RRT remains orders of magnitude cheaper than population training in both tokens and time at low N ; its cost rises with dimension because each edge check needs more queries and the free fraction shrinks. The Halton–prime sampler is deterministic and low-discrepancy (12); its advantage over uniform sampling in iterations and collision queries tends to appear, and to widen, at higher DOF — exactly where coverage quality matters most, because uncovered corridors are what starve a bidirectional tree.

Accuracy versus generality. Method B reaches the goal more precisely (the DLS refinement of (22)–(25) pins the endpoint) but guarantees only endpoints and collision-freeness of one path. Method A explicitly optimizes Cartesian tracking along the *whole* path (15) and amortizes its training cost when the trained policy is reused across queries — the classical amortization argument for learned IK, here verified to survive the sweep from 5 to 30 DOF.

The prime lens, summarized. The two methods are usually presented as rivals; the prime principle reveals a symmetry between them. Both need, per degree of freedom, an integer label whose interactions with all other labels are as “generic” as integer arithmetic allows — no shared harmonics for the encoder, no shared lattice cycles for the sampler. Pairwise coprimality is that genericity, and the primes deliver it canonically, deterministically, and inexhaustibly. Increasing the number of degrees of freedom then stops being a representational event at all: it is the consumption of the next element of a sequence that number theory finished preparing long ago.

8 Conclusion

We presented a matched pair of motion-generation pipelines for planar redundant arms — a prime-frequency-encoded neural population and a Halton–prime configuration-space planner — evaluated under a constant-reach protocol that isolates dimensionality from 5 to 30 DOF. All kinematic, collision, loss, and planning equations of the accompanying experiment suite were given in closed form, together with the robustness machinery required at high DOF. Beyond the specific systems,

the paper argues for a design stance: when a representation must grow with the number of degrees of freedom, indexing those degrees by successive primes turns scaling into appending — an elegant, deterministic, and arithmetically guaranteed alternative to re-engineering the representation at every new joint. Natural extensions include scrambled or leaped Halton variants for very high dimensions, spatial (3-D) arms where self-collision geometry is richer, and learned samplers whose priors are seeded by, rather than replaced by, the prime construction.

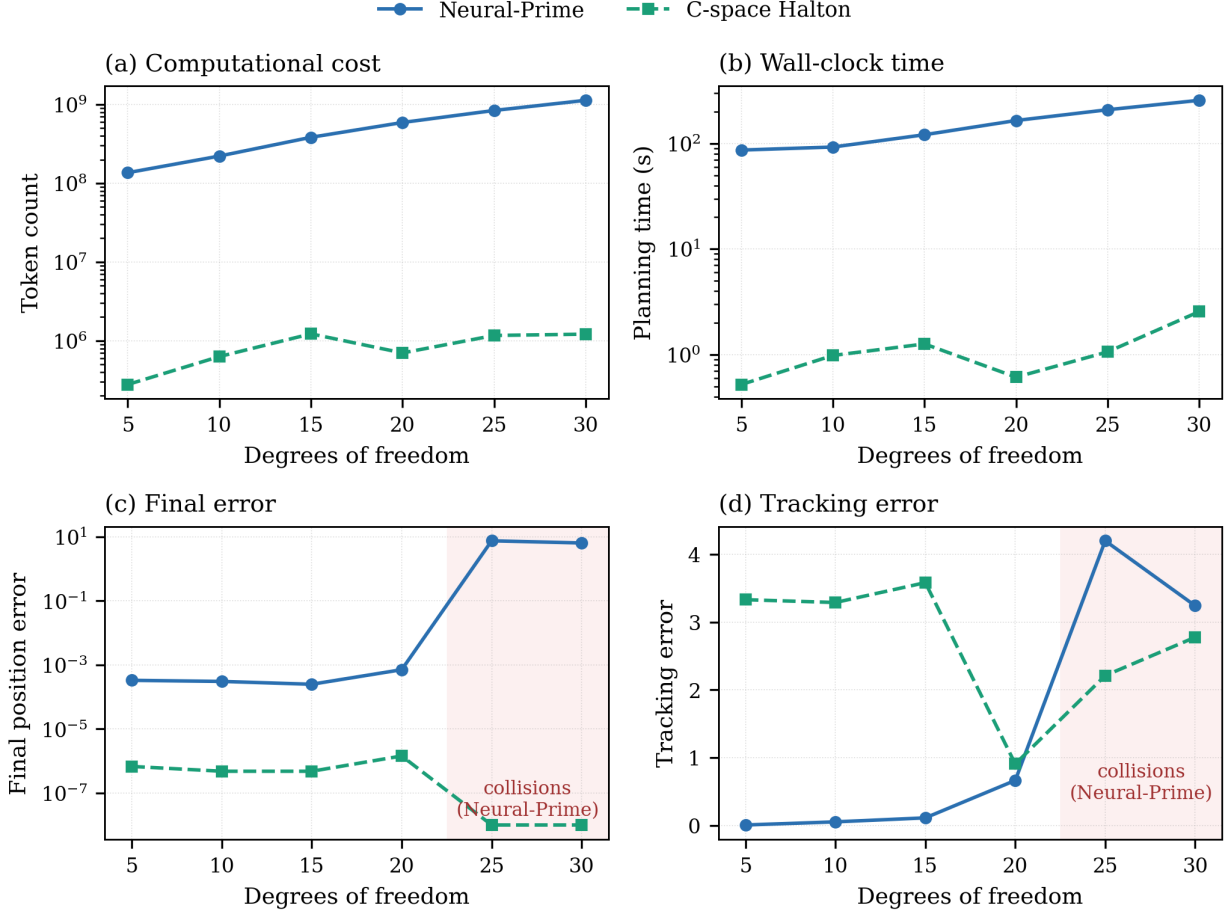


Figure 1: Comparison between Neural-Prime and C-space Halton for planning problems with different numbers of degrees of freedom. Neural-Prime exhibits substantially higher computational cost and wall-clock time, whereas C-space Halton maintains lower final-position errors. At higher dimensionalities, Neural-Prime also experiences collisions and a significant increase in both final and tracking errors. [This and other plots and notebook are available here.](#)

The results presented here acquire their full significance when projected toward systems with a large number of degrees of freedom. In the low-dimensional case studied, GPU-based trajectory optimisation already demonstrates that the per-iteration cost of resolving coupled, energy-aware dynamics can be brought within real-time budgets; as the dimensionality grows — from a single actuated suspension to legged platforms, dual-arm systems, and fully dexterous hands — the matrix structure of the underlying computations (batched linearisations, Riccati-like recursions, and parallel rollout evaluation) scales naturally onto massively parallel hardware, whereas CPU-bound

implementations degrade rapidly.

A particularly compelling long-term perspective is the migration of this computation from a single centralised processor toward the mechanical structure itself: a future in which each joint carries its own embedded GPU or GPU-class accelerator. Under such an architecture, every articulation becomes a computational node capable of locally evaluating its own dynamics, contact conditions, and energy exchange, while a lightweight consensus or message-passing layer coordinates the global objective. This turns whole-body control into a distributed optimisation problem whose factor-graph structure mirrors the kinematic chain, and it offers three concrete advantages: (i) latency between sensing and actuation collapses to the local loop, enabling reflex-like responses to the abrupt, localised disturbances discussed above; (ii) the system degrades gracefully, since the failure of one computational node does not silence the entire controller; and (iii) the energy-aware formulation developed here becomes local by construction, with each joint managing its own dissipation budget. The methods and results of this work should therefore be read not only as a solution for the present low-dimensional benchmark, but as a computational template deliberately chosen to remain valid when the controller is replicated across dozens of articulations, each endowed with its own parallel processor.

References

- [1] J. H. Halton. On the efficiency of certain quasi-random sequences of points in evaluating multi-dimensional integrals. *Numerische Mathematik*, 2:84–90, 1960.
- [2] H. Niederreiter. *Random Number Generation and Quasi-Monte Carlo Methods*. SIAM, 1992.
- [3] S. M. LaValle and J. J. Kuffner. Randomized kinodynamic planning. *International Journal of Robotics Research*, 20(5):378–400, 2001.
- [4] J. J. Kuffner and S. M. LaValle. RRT-Connect: An efficient approach to single-query path planning. In *Proc. IEEE ICRA*, 2000.
- [5] Y. Nakamura and H. Hanafusa. Inverse kinematic solutions with singularity robustness for robot manipulator control. *ASME J. Dynamic Systems, Measurement, and Control*, 108:163–171, 1986.
- [6] C. W. Wampler. Manipulator inverse kinematic solutions based on vector formulations and damped least-squares methods. *IEEE Trans. Systems, Man, and Cybernetics*, 16(1):93–101, 1986.
- [7] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In *Proc. ECCV*, 2020.
- [8] M. Tancik, P. P. Srinivasan, B. Mildenhall, et al. Fourier features let networks learn high frequency functions in low dimensional domains. In *Proc. NeurIPS*, 2020.